

Lines and Planes in Three Dimensions

ANSWER KEY



The vector equation of a line

- 1 Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.
 $\mathbf{r} = (2, -1, 3) + t(1, 2, -1)$, where $t \in \mathbb{R}$.
- 2 Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.
Direction vector: $\vec{AB} = (4 - 1, 3 - 0, 1 - 2) = (3, 3, -1)$. Vector equation: $\mathbf{r} = (1, 0, 2) + t(3, 3, -1)$.

Determining if two lines intersect

- 3 Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.
Equate components: $1 + s = 2t$, $2 - s = 3 + t$, $2s = 1 - t$. From the second: $s = -1 - t$. Substitute into the first: $1 + (-1 - t) = 2t$, so $-t = 2t$, giving $t = 0$, then $s = -1$. Check the third equation: $2(-1) = 1 - 0$? $-2 = 1$ — false. The lines do not intersect (they are skew, since they are not parallel either).

The equation of a plane

- 4 Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.
The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$.
Equation: $2x - y + 3z = -3$.
- 5 Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.
 $\vec{AB} = (-1, 1, 0)$, $\vec{AC} = (-1, 0, 1)$. Normal
 $= \vec{AB} \times \vec{AC} = (1 \cdot 1 - 0 \cdot 0, 0 \cdot (-1) - (-1) \cdot 1, (-1) \cdot 0 - 1 \cdot (-1)) = (1, 1, 1)$. Plane: $x + y + z = d$;
substituting A: $1 + 0 + 0 = 1$. Equation: $x + y + z = 1$.

Angle between a line and a plane

- 6 Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.
Plane normal: $\mathbf{n} = (1, 1, 1)$. $\mathbf{d} \cdot \mathbf{n} = 1 + 2 + 2 = 5$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = \sqrt{3}$.
 $\sin \theta = |\mathbf{d} \cdot \mathbf{n}| / (|\mathbf{d}| |\mathbf{n}|) = 5 / (3\sqrt{3}) \approx 0.962$, so $\theta \approx 74^\circ$.

Distance from a point to a plane

- 7 Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.
- $$d = |2(3) - 1(1) + 2(-2) - 6|/\sqrt{4 + 1 + 4} = |6 - 1 - 4 - 6|/3 = |-5|/3 = 5/3.$$

Finding the angle between two planes

- 8 Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.
- Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$.
- $$\cos \theta = |-1|/(\sqrt{6} \cdot \sqrt{6}) = 1/6, \text{ so } \theta \approx 80^\circ.$$

Finding the intersection line of two planes

- 9 Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.
- Direction:
- $$\mathbf{n}_1 \times \mathbf{n}_2 = (1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1), 1 \cdot 1 - 1 \cdot 2, 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2).$$
- To find a point, try $z = 0$: $x + y = 6$ and $x - y = 5$, giving $x = 5.5$, $y = 0.5$. Point: $(5.5, 0.5, 0)$. Line:
- $$\mathbf{r} = (5.5, 0.5, 0) + t(3, -1, -2).$$

Testing whether a line lies within a plane

- 10 Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.
- A line lies within a plane if (1) its direction vector is perpendicular to the plane's normal, and (2) a point on the line satisfies the plane's equation. Normal of the plane: $(1, 3, 1)$. Direction of line: $(2, -1, 1)$. Dot product: $1(2) + 3(-1) + 1(1) = 2 - 3 + 1 = 0$ ✓ (perpendicular to normal, so parallel to the plane). Check the point $(1, 2, 3)$: $1 + 3(2) + 3 = 1 + 6 + 3 = 10$ ✓ (satisfies the plane equation). Since both conditions hold, the line lies entirely within the plane.

Finding the point where a line meets a plane

- 11 Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.
- Substitute the parametric coordinates $x = 1 + t$, $y = 2t$, $z = 2 - t$ into the plane equation:
- $$2(1 + t) + 2t - (2 - t) = 3. \text{ Expand: } 2 + 2t + 2t - 2 + t = 3, \text{ so } 5t = 3, \text{ giving } t = 3/5. \text{ Substituting back: } x = 1 + 3/5 = 8/5, y = 2(3/5) = 6/5, z = 2 - 3/5 = 7/5. \text{ Point of intersection: } (8/5, 6/5, 7/5).$$

Shortest distance between two skew lines

- 12** Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)| / |\mathbf{d}_1 \times \mathbf{d}_2|$.
- $\mathbf{d}_1 \times \mathbf{d}_2 = (0 \cdot 1 - 1 \cdot 1, 1 \cdot 0 - 1 \cdot 1, 1 \cdot 1 - 0 \cdot 0) = (-1, -1, 1)$. $\vec{AB} = (1, 1, 0)$.
- $\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) = 1(-1) + 1(-1) + 0(1) = -2$. $|\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{1 + 1 + 1} = \sqrt{3}$. Distance $= |-2|/\sqrt{3} = 2/\sqrt{3} = 2\sqrt{3}/3$.
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Foot of the perpendicular from a point to a line

- 13** Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.
- A general point on the line is $(1 + t, t, t)$. The vector from this point to P is $(4 - (1 + t), 0 - t, 1 - t) = (3 - t, -t, 1 - t)$. For the foot of the perpendicular, this vector must be perpendicular to the direction $(1, 1, 1)$: $(3 - t)(1) + (-t)(1) + (1 - t)(1) = 0$, i.e. $3 - t - t + 1 - t = 0$, so $4 - 3t = 0$, giving $t = 4/3$. Foot of perpendicular: $(1 + 4/3, 4/3, 4/3) = (7/3, 4/3, 4/3)$.
- 14** Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.
- The second equation's coefficients $(-4, 2, -6)$ are exactly -2 times the first equation's coefficients $(2, -1, 3)$, and the constant -8 is also -2 times 4 . Since the entire equation is a scalar multiple of the first, the two equations describe the same plane.
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Checking whether a point lies on a plane

- 15** Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.
- Substituting P : $3(2) + 2(-1) - 4 = 6 - 2 - 4 = 0 \neq 1$, so P does not lie on the plane. Substituting the origin $(0, 0, 0)$: $3(0) + 2(0) - 0 = 0$. Since the plane value at P (0) equals the plane value at the origin (0), and both are less than the required value of 1 , P lies on the same side of the plane as the origin.
- 16** Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.
- A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector. Normal of the plane: $(1, 2, -1)$. Direction of line: $(2, -1, k)$. Setting the dot product to zero: $1(2) + 2(-1) + (-1)(k) = 0$, so $2 - 2 - k = 0$, giving $k = 0$.
- 17** A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z -axis (i.e. the height at $x = 0, y = 0$).
- Substituting $x = 6, y = 4$: $2(6) + 3(4) + 6z = 42$, so $12 + 12 + 6z = 42$, giving $6z = 18$, so $z = 3$ metres. The z -intercept (at $x = 0, y = 0$): $6z = 42$, so $z = 7$ metres.

- 18** This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.
- (a) Both lines have z -component fixed at 1 (since the direction vectors $(1, 1, 0)$ and $(1, -1, 0)$ have zero z -component, and both starting points have $z = 1$), so every point on either line satisfies $z = 1$; both lines lie in the plane $z = 1$. (b) Equating: $2 + s = t$ and $s = 2 - t$. Substituting the second into the first: $2 + (2 - t) = t$, so $4 - t = t$, giving $t = 2$, then $s = 0$. Point: $\mathbf{r}_1$ at $s = 0$ gives $(2, 0, 1)$; checking $\mathbf{r}_2$ at $t = 2$: $(0, 2, 1) + 2(1, -1, 0) = (2, 0, 1)$ ✓. Intersection point: $(2, 0, 1)$. (c) Direction vectors: $(1, 1, 0)$ and $(1, -1, 0)$. Dot product: $1 - 1 + 0 = 0$. Since the dot product is zero, the lines are perpendicular, so the angle between them is 90° .

Converting between vector and Cartesian forms of a line

- 19** Write the parametric equations of the line $\mathbf{r} = (2, 1, -3) + t(1, -2, 4)$, and hence give its Cartesian (symmetric) form.
- Parametric: $x = 2 + t, y = 1 - 2t, z = -3 + 4t$. Solving each for t : $t = x - 2, t = (1 - y)/2, t = (z + 3)/4$.
 Cartesian form: $(x - 2)/1 = (1 - y)/2 = (z + 3)/4$.

Determining if two planes are parallel

- 20** Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.
- The second normal $(2, -4, 6)$ is exactly 2 times the first normal $(1, -2, 3)$, so the planes are parallel. If they were the same plane, the constant term would also scale by 2: $2 \times 5 = 10$. Since the actual constant is $1 \neq 10$, the planes are parallel but distinct.

Distance between two parallel planes

- 21** Find the distance between the parallel planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ from the previous question.
- Divide the second equation by 2 so both equations share the same normal: $x - 2y + 3z = 0.5$. Using the point-to-plane distance formula with any point on the first plane (or directly comparing constants), the distance is $|5 - 0.5|/\sqrt{1^2 + (-2)^2 + 3^2} = 4.5/\sqrt{14}$.

Checking whether three points are collinear

- 22** Determine whether the points $A(1, 2, 3)$, $B(2, 4, 6)$, and $C(4, 8, 12)$ are collinear.
- $\vec{AB} = (2 - 1, 4 - 2, 6 - 3) = (1, 2, 3)$. $\vec{AC} = (4 - 1, 8 - 2, 12 - 3) = (3, 6, 9)$. Since $\vec{AC} = 3\vec{AB}$, the two vectors are parallel and share the common point A , so A , B , and C are collinear.

Reflecting a point in a plane

23 Find the reflection of the point $P(1, 2, 0)$ in the plane $x + y + z = 6$.

The line through P along the plane's normal is $\mathbf{r} = (1, 2, 0) + t(1, 1, 1)$. Substituting into the plane equation: $(1 + t) + (2 + t) + (0 + t) = 6$, so $3 + 3t = 6$, giving $t = 1$. The foot of the perpendicular is $(2, 3, 1)$. The reflection of P is twice the foot minus P :

$$2(2, 3, 1) - (1, 2, 0) = (4, 6, 2) - (1, 2, 0) = (3, 4, 2).$$